

Activity - 2

Stack, Queue, Linked List

29/7/26

→ $10\ 5 + 60\ 6 / * 8 -$ is

1) Stack = $54 \rightarrow 54, 52 \rightarrow \text{pop}(52) \rightarrow 54, 55 \rightarrow$
 $54, 55, 62 \rightarrow S = 62.$

Queue = $21 \rightarrow 21, 24 \rightarrow \text{Dequeue}(21) \rightarrow$

$24, 28 \rightarrow 24, 28, 32 \rightarrow q = 24$

$$S + q = 62 + 24 = 86.$$

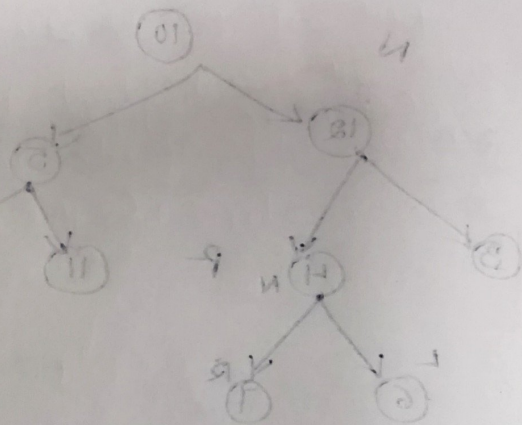
2) $10\ 5 + 60\ 6 / * 8 -$

$$10 + 5 = 15$$

$$60 \div 6 = 10$$

$$15 \times 10 = 150$$

$$150 - 8 = 142$$



3) Reverse Queue.

move elements between Q_1 & Q_2 until Q_2 becomes $4, 3, 2, 1$

Answer: 14 enqueue operations.

4) Reverse stack.

Statement II is true.

5) Postfix expression.

After the first *, the top two stack elements are 2, 18. 6, 1

6) Infix $\rightarrow$ Postfix

$$a + b * c - d ^ 1 e ^ 1 f$$

Postfix = $abc * def ^ 1 ^ 1 -$

7) Stack operations

Popped values.

20 $\rightarrow$ 20 $\rightarrow$ 10 $\rightarrow$ 10 $\rightarrow$ 20.